Incidence Matrix

1 Definition: Incidence Matrix

Let V be a finite set of n elements (often called vertices, points, or elements), and let $\mathcal{F} = \{S_1, S_2, \ldots, S_m\}$ be a family of m subsets of V.

The **incidence matrix** of the set system $(V, \mathcal{F})$ is an $m \times n$ matrix A whose rows are indexed by the sets S_i and whose columns are indexed by the elements $v_j \in V$.

The entries of the matrix, $a_{i,j}$, are defined as:

$$a_{i,j} = \begin{cases} 1 & \text{if } v_j \in S_i \\ 0 & \text{if } v_j \notin S_i \end{cases}$$

In other words, a 1 in row i and column j indicates that the element v_j is *incident* to (contained in) the set S_i. (Note: In some algebraic graph theory contexts, entries can be -1 to denote direction, but in discrepancy theory and set systems, they are strictly 0 and 1).

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2 Concrete Example

Let $V = \{a, b, c, d\}$ be a ground set of 4 elements. Let $\mathcal{F}$ be a set system containing 3 subsets:

- $S_1 = \{a, b, c\}$
- $S_2 = \{b, d\}$
- $S_3 = \{a, c, d\}$

To build the incidence matrix A, we create 3 rows (for the sets) and 4 columns (for the elements a, b, c, d in order):

$$A = \begin{array}{c} \\ S_1 \\ S_2 \\ S_3 \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix} \end{array}$$

- Look at Row 1 (S_1): It has 1s in columns a, b, and c, because those elements are in S_1. It has a 0 in column d.
- Look at Row 2 (S_2): It has 1s in columns b and d.
- Look at Row 3 (S_3): It has 1s in columns a, c, and d.

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3 How the Incidence Matrix Works in Discrepancy Theory

The incidence matrix allows us to turn combinatorial set operations into linear algebra.

Let χ be a coloring vector. If we have elements $V = \{a, b, c, d\}$, the coloring is a column vector:

$$\mathbf{x} = \begin{pmatrix} \chi(a) \\ \chi(b) \\ \chi(c) \\ \chi(d) \end{pmatrix} \in \{-1, 1\}^4$$

Now, let's multiply the incidence matrix A by the coloring vector $\mathbf{x}$:

$$A\mathbf{x} = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} \chi(a) \\ \chi(b) \\ \chi(c) \\ \chi(d) \end{pmatrix} = \begin{pmatrix} \chi(a) + \chi(b) + \chi(c) \\ \chi(b) + \chi(d) \\ \chi(a) + \chi(c) + \chi(d) \end{pmatrix}$$

Notice what happened: **The i-th entry of the resulting vector $A\mathbf{x}$ is exactly the sum of the colors of the elements inside the set S_i.**

Therefore, the discrepancy vector (the imbalances of all the sets) is simply the matrix-vector product $A\mathbf{x}$. The problem of finding a coloring that minimizes discrepancy is algebraically characterized as finding a vector $\mathbf{x} \in \{-1, 1\}^n$ that minimizes the norm of $A\mathbf{x}$.